

The Pais–Uhlenbeck Metric as a Minimum-Distortion Principle, and the Representation Problem for the Fourth-Order Field

Abstract

For the unequal-frequency PT-symmetric Pais–Uhlenbeck oscillator, the admissible complex-symplectic diagonalizers form a coset $S_+ SO(2, \mathbb{C})^2$ of the unique Hermitian positive-definite diagonalizer S_+ . We prove that S_+ minimizes the affine-invariant metric distortion: for every admissible S , $\|\log(S^\dagger S)\|_F^2 \geq 4r^2$ with $r = \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}$, with equality exactly on the unitary stabilizer coset, on which the induced Hilbert-space metric is constantly the Bender–Mannheim $\eta = e^{-Q}$. The proof is elementary and exact: at worst-case stabilizer alignment the log-eigenvalues of $S^\dagger S$ are $\operatorname{arccosh}(\cosh r \cosh b) \pm a$ in the Gram rapidities (a, b) of the stabilizer factor. The general result behind it is an orthogonal Cartan-norm projection principle, proved in if-and-only-if form, whose principal case is the $\operatorname{Sp}(2n, \mathbb{C})$ inter-mode theorem: relative to any symplectic mode splitting, minimality along the mode-preserving coset holds exactly for inter-mode diagonalizer directions. The Pais–Uhlenbeck oscillator is the fully integrable $n = 2$ model: the physical stabilizer, itself not θ -stable, has canonical compatible hull exactly the mode-preserving $SL(2, \mathbb{C})^2$ (totally geodesic $\mathbb{H}^3 \times \mathbb{H}^3$), the Bender–Mannheim direction is inter-mode, and the closed form above is the resulting mixed hyperbolic/flat Pythagorean law. We then quantize the free fourth-order field modewise and compute the Bender–Mannheim ground state exactly: it is a normalizable real Gaussian for every mode, but its fidelity with the two-field Fock vacuum tends to the universal constant $\sqrt{3}/2$ at large momentum, and its relative particle content to 1/3 per mode pair. Consequently the physical (PT) and naive Fock representations of the free fourth-order field are disjoint in every spatial dimension, although each mode is unitarily equivalent: PT unitarization is exact modewise and changes the field-theoretic representation. All identities are machine-verified.

1 Introduction

This paper continues the symplectic analysis of the PT-symmetric Pais–Uhlenbeck (PU) oscillator begun in [1], whose results and conventions we use freely: $V = (x, y, p, q)^\top$, $[V_a, V_b] = iJ_{ab}$, $H_{\text{PT}} = \frac{1}{2}V^\top G V$ with the PU matrix G , positive normal form G_0 , rapidity

$$r = \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2}, \quad \omega_1 > \omega_2 > 0, \quad (1)$$

and, after the canonical normalization $d_x d_y = \gamma \omega_1 \omega_2$, the unique Hermitian positive-definite diagonalizer

$$S_+ = \cosh \frac{r}{2} I + \sinh \frac{r}{2} B \in \operatorname{Sp}(4, \mathbb{C}), \quad S_+^\top G' S_+ = G'_0, \quad S_+^\top J S_+ = J, \quad (2)$$

with B a fixed Hermitian involution. The full solution family of the diagonalization problem is the coset $S_+ \operatorname{Stab}(J, G'_0)$ with $\operatorname{Stab}(J, G'_0) \cong SO(2, \mathbb{C}) \times SO(2, \mathbb{C})$, and the metaplectic logarithm of S_+ is the Bender–Mannheim generator $Q = \alpha p q + \beta x y$ [2, 3, 1].

Two questions are addressed here.

First, uniqueness of the positive diagonalizer does not by itself distinguish S_+ variationally: one may ask whether S_+ *minimizes* a natural functional over all admissible diagonalizers. We prove that it minimizes the affine-invariant distortion $\mathcal{F}(S) = d(I, S^\dagger S)^2 = \|\log(S^\dagger S)\|_F^2$ (Theorem 2.5), so that the Bender–Mannheim metric is selected by a minimum-distortion principle: it is the least-distorting complex canonical transformation converting the PU Hamiltonian to its positive normal form. The results of Section 2 form a hierarchy:

normal-coset projection principle (Thm. 2.8) $\implies$ $\mathrm{Sp}(2n, \mathbb{C})$ inter-mode theorem (Thm. 2.13) $\implies$ exact $\mathbb{H}^2 \times \mathbb{H}^2$

The first is classical symmetric-space geometry, stated here in if-and-only-if form; the second is the principal general result, making the selection principle a theorem for arbitrary quadratic PT mode systems; the third is what only the integrable $n = 2$ case supplies — the exact minimizing value, obtained by the elementary two-invariant derivation that opens the section. The physical stabilizer itself fails θ -stability — which is why the classical theory does not apply to it directly — but its canonical compatible hull is exactly the mode-preserving $SL(2, \mathbb{C})^2$ (Proposition 2.9, stratified over the alignment locus).

Second, for the free fourth-order field $(\square + m_1^2)(\square + m_2^2)\phi = 0$ each spatial momentum k contributes a PU pair with $\omega_i(k) = \sqrt{k^2 + m_i^2}$, and the modewise construction applies with rapidity

$$r(k) = \log \frac{(\omega_1(k) + \omega_2(k))^2}{m_1^2 - m_2^2} = 2 \log |k| + O(1) \quad (m_1 > m_2). \quad (3)$$

One might expect from (3) that the modewise transformation carries unboundedly growing squeezing and fails Fock implementability “dramatically”. The exact computation shows something sharper and more structured (Theorems 3.2–3.4): the PT ground state exists as a normalizable Gaussian for *every* mode, its mismatch with the free two-field vacuum saturates at *universal constants* — fidelity $\sqrt{3}/2$ and relative occupation $1/3$ per mode pair at large k — and therefore the physical and naive representations are disjoint in every spatial dimension $d \geq 1$, with vacuum overlap decaying like $e^{-cV\Lambda^d}$ under a cutoff. The divergence of $r(k)$ itself is a statement about the observable-space (phase-space) metric, which we separate cleanly (Corollary 3.1): its eigenvalues $e^{\pm r(k)} \asymp (1 + k^2)^{\pm 1}$ make the phase-space metric an $H^1 \oplus H^{-1}$ -type Sobolev pair, unboundedly inequivalent to the standard norm, while the state-level mismatch remains $O(1)$ per mode.

Verification. As in [1], every finite-dimensional identity in this paper — the invariant formulas, the closed-form log-eigenvalues, the PT ground-state Gaussian, the eigenvalue equation, the fidelity and occupation formulas, and the limits $\sqrt{3}/2$ and $1/3$ — is machine-verified symbolically, with independent numerical checks (global optimization for Theorem 2.5; high-precision quadrature for Theorem 3.2). The scripts accompany the verification repository of [1].

2 The minimum-distortion theorem

Work in the canonically normalized coordinates of [1]; all adjoints refer to them. The stabilizer is parametrized by two complex angles,

$$C(\theta_1, \theta_2) = \sum_{j=1,2} (\cos \theta_j P_j + \sin \theta_j X_j), \quad X_j^2 = -P_j, \quad (4)$$

with P_j the projections onto the modes of G'_0 and X_j the per-mode complex structures; every admissible diagonalizer is $S = S_+ C$.

Definition 2.1. For an admissible S let $\mathcal{F}(S) = \|\log(S^\dagger S)\|_F^2$, the squared affine-invariant distance of the observable-space metric $S^\dagger S$ from the identity [4]. For a stabilizer element $C = C_1 \oplus C_2$ (blocks on the two modes) define its *Gram rapidities* $\tau_j \geq 0$ by

$$W_j = C_j C_j^\dagger, \quad \det W_j = 1, \quad \cosh \tau_j = \frac{1}{2} \operatorname{tr} W_j, \quad (5)$$

and set $a = \frac{1}{2}(\tau_1 + \tau_2)$, $b = \frac{1}{2}(\tau_1 - \tau_2)$.

Note $\tau_1 = \tau_2 = 0$ iff $W_1 = W_2 = I$ iff C is unitary. Since $P = S^\dagger S = C^\dagger S_+^2 C$ is Hermitian positive and symplectic, its eigenvalues form reciprocal pairs $e^{\pm x_1}, e^{\pm x_2}$ with $x_i \geq 0$, and $\mathcal{F}(S) = 2(x_1^2 + x_2^2)$.

Lemma 2.2 (Invariants). *Write $W = C C^\dagger = W_1 \oplus W_2$, $c = \cosh r$, $s = \sinh r$. Decomposing $W_j = \cosh \tau_j I + \sinh \tau_j (n_j \cdot \sigma)$ with real unit vectors n_j , and letting $\chi \in [-1, 1]$ denote the alignment $n_1 \cdot \tilde{n}_2$ (where $\tilde{n}_2$ is n_2 with its σ_2 component reflected),*

$$\cosh x_1 + \cosh x_2 = c (\cosh \tau_1 + \cosh \tau_2), \quad (6)$$

$$\cosh x_1 \cosh x_2 = \frac{1}{2} \left[(2c^2 - s^2) \cosh \tau_1 \cosh \tau_2 + s^2 (1 + \sinh \tau_1 \sinh \tau_2 \chi) \right]. \quad (7)$$

Proof. $\operatorname{tr} P = \operatorname{tr}(e^{rB} W) = c \operatorname{tr} W + s \operatorname{tr}(BW)$, and $\operatorname{tr}(BW) = 0$ because B is off-diagonal and W diagonal with respect to the mode splitting; this gives (6). For (7), expand $\operatorname{tr} P^2 = c^2 \operatorname{tr} W^2 + s^2 \operatorname{tr}((BW)^2)$ (the cross terms again vanish blockwise); the off-diagonal blocks of B equal $-\sigma_2$, and the 2×2 identity $\sigma_2 M \sigma_2 = (\det M) M^{-\top}$ reduces $\operatorname{tr}((BW)^2) = 2 \operatorname{tr}(\overline{W_2}^{-1} W_1) = 4[\cosh \tau_1 \cosh \tau_2 - \sinh \tau_1 \sinh \tau_2 \chi]$. Converting $(\operatorname{tr} P, \operatorname{tr} P^2)$ to the symmetric functions of $\cosh x_i$ yields (7). $\square$

(As verification evidence, both matrix identities are also checked against random stabilizer elements in the repository.)

Lemma 2.3 (Worst alignment; closed form). *At $\chi = -1$ the system (6)–(7) is solved exactly by*

$$x_{1,2} = \varphi(b) \pm a, \quad \varphi(b) = \operatorname{arccosh}(\cosh r \cosh b) \geq r. \quad (8)$$

Proof. With $\alpha = \cosh a$, $\beta = \cosh b$ one finds from (6)–(7) at $\chi = -1$, using $\cosh \tau_1 \cosh \tau_2 \pm \sinh \tau_1 \sinh \tau_2 = \cosh(\tau_1 \pm \tau_2)$,

$$\cosh x_1 + \cosh x_2 = 2c\alpha\beta, \quad \cosh x_1 \cosh x_2 = \alpha^2 + c^2\beta^2 - 1,$$

whence $\cosh x_{1,2} = c\alpha\beta \pm \sqrt{(\alpha^2 - 1)(c^2\beta^2 - 1)} = \cosh(\varphi \pm a)$ with $\cosh \varphi = c\beta$, by the hyperbolic addition formulas (elementary identities; also machine-checked in the repository as verification evidence). Finally $\cosh \varphi(b) - \cosh r = \cosh r(\cosh b - 1) \geq 0$ gives $\varphi(b) \geq r$ with equality iff $b = 0$. $\square$

Lemma 2.4 (Alignment monotonicity). *At fixed $\cosh x_1 + \cosh x_2$, the quantity $x_1^2 + x_2^2$ is nondecreasing in $\cosh x_1 \cosh x_2$; and (7) is nondecreasing in χ .*

Proof. Parametrize $\cosh x_{1,2} = T/2 \pm w$, $w \geq 0$; the product decreases in w , and $\frac{d}{dw}(x_1^2 + x_2^2) = 2(\frac{x_1}{\sinh x_1} - \frac{x_2}{\sinh x_2}) \leq 0$ because $x/\sinh x$ is strictly decreasing and $x_1 \geq x_2$. The second statement is the sign of the coefficient $s^2 \sinh \tau_1 \sinh \tau_2 / 2 \geq 0$ in (7). $\square$

Theorem 2.5 (Minimum distortion). *For every admissible diagonalizer $S = S_+ C$,*

$$\mathcal{F}(S) \geq 4 \left[a^2 + \operatorname{arccosh}^2 (\cosh r \cosh b) \right] \geq 4r^2 = \mathcal{F}(S_+), \quad (9)$$

with overall equality if and only if C is unitary. All minimizers share the polar factor S_+ , hence the same Hilbert-space metric $\eta = e^{-Q}$: the Bender–Mannheim metric is the unique metric of any minimum-distortion diagonalizer,

$$S_+ \in \arg \min_{S^\top JS = J, S^\top G' S = G'_0} d(I, S^\dagger S), \quad Q = -2 \log (\text{polar factor of any minimizer})|_{\text{metaplectic}}. \quad (10)$$

Proof. By Lemma 2.4, lowering χ to -1 at fixed Gram rapidities does not increase $x_1^2 + x_2^2$; by Lemma 2.3 the value there is $\mathcal{F} = 2[(\varphi + a)^2 + (\varphi - a)^2] = 4(\varphi^2 + a^2)$; and $\varphi(b) \geq r$. Equality in the chain forces $a = 0$ and $b = 0$, i.e. $\tau_1 = \tau_2 = 0$, i.e. C unitary; the converse is immediate since then $S^\dagger S$ is unitarily conjugate to S_+^2 . If C is unitary, $S_+ C = S_+ \cdot C$ is already a polar decomposition, so the polar factor is S_+ and the metaplectic implementer factorizes as $\rho' = U_C e^{-Q/2}$ with U_C unitary, giving $\eta' = \rho'^\dagger \rho' = e^{-Q}$ for every minimizer. $\square$

Remark 2.6. (i) The individual log-eigenvalues do *not* satisfy $x_i \geq r$: numerical scans show $\min_i x_i < r$ on open regions of the stabilizer; only the joint quantity is bounded below, which is why the two-invariant argument is needed. (ii) The equality set is the unitary part of the stabilizer, which in the canonical coordinates is generically the finite group $\mathbb{Z}_2 \times \mathbb{Z}_2$ [1]; abstractly it is contained in the maximal compact $U(1)^2$ of $SO(2, \mathbb{C})^2$. (iii) The theorem is an instance of an orthogonal Cartan-norm projection principle, proved in Section 2.1 together with its converse; the elementary proof above remains valuable because it yields the exact value (8), which the geometric argument does not.

2.1 Geometric recognition: an orthogonal Cartan-norm projection principle

Normalization dictionary. Define the Cartan projection $\mu(S)$ of $S \in \operatorname{Sp}(4, \mathbb{C})$ as the chamber-ordered log-spectrum of $S^\dagger S$ (twice the usual log-singular-value vector), equip the Hermitian directions $\mathfrak{p}$ with the trace form $\langle \xi, \eta \rangle = \operatorname{tr}(\xi \eta)$, and normalize the symmetric-space distance by $d_X(o, e^\xi \cdot o) = \|\xi\|_F$. If $\operatorname{spec}(\xi) = \{\pm a_1, \pm a_2\}$ then $\mu(e^\xi) = (2a_1, 2a_2)$ (chamber-ordered), $\|\xi\|_F^2 = 2 \sum_i a_i^2$, and the three minimands used below are related by

$$\boxed{\mathcal{F}(g) = 2\|\mu(g)\|^2, \quad \mathcal{F}(e^\xi) = 4d_X(o, e^\xi \cdot o)^2 = \|2\xi\|_F^2.} \quad (11)$$

In particular $\min \sqrt{\mathcal{F}} = \|2\xi\|_F$, $\min \|\mu\| = \|2\xi\|_F / \sqrt{2}$, and the first-variation coefficient 8 in (12) below is consistent with $\mathcal{F}(e^\xi) = 4\|\xi\|_F^2$. Throughout, K denotes the maximal compact subgroup determined by the physical inner product: $K = \operatorname{Sp}(4, \mathbb{C}) \cap U(4) \cong \operatorname{Sp}(2)$ (the compact symplectic group, written $USp(4)$ in part of the literature), and in the n -mode setting $K = \operatorname{Sp}(2n, \mathbb{C}) \cap U(2n) \cong \operatorname{Sp}(n)$. With these conventions $\mu(S_+) = (r, r)$, which lies *on the chamber wall* $x_1 = x_2$ of type C_2 .

Theorem 2.5 is thus a minimization of the Cartan-projection *norm* along a stabilizer coset. We emphasize at the outset what is classical and what is not. The scalar projection inequality below is standard symmetric-space geometry in the tradition of Mostow's self-adjoint groups [5, 7]: the compatible-subgroup orbit is totally geodesic and convex, and normal geodesics realize nearest-point projection. (Kempf–Ness theory [6] offers a related variational perspective on minimal-norm points of complexified orbits, but it is not the direct source of this coset inequality.) The inequality controls $\|\mu\|$, not the vector-valued Cartan projection in Weyl-chamber order, and we do not claim it as new. The structure specific to this problem is: (i) the physically given stabilizer $H = \text{Stab}(J, G'_0)$ is *not* θ -stable, so the classical theory does not see it directly; (ii) its canonical θ -stable hull is exactly the mode-preserving $SL(2, \mathbb{C}) \times SL(2, \mathbb{C})$ (Proposition 2.9), through which minimality is inherited by restriction; and (iii) the exact $\mathbb{H}^2 \times \mathbb{R}$ value formula (8).

Let $\mathbb{P}$ denote the cone of positive-definite Hermitian matrices with the affine-invariant metric, a nonpositively curved (NPC) space in which $t \mapsto d(\gamma(t), x)^2$ is strictly convex along nonconstant geodesics, with $\frac{d}{dt} d(e^{t\zeta}, P)^2|_{t=0} = -2\langle \zeta, \log P \rangle$ [4, 7].

Lemma 2.7. *Let $\mathcal{C} \subset GL(n, \mathbb{C})$ be a closed subgroup, closed under conjugate transpose, and let $\mathbb{P}_{\mathcal{C}} = \mathcal{C} \cap \mathbb{P}$ be its set of positive elements. Then $\mathbb{P}_{\mathcal{C}}$ is a topologically closed, geodesically closed (hence convex) subset of $\mathbb{P}$ — so it admits unique metric projection from any point of $\mathbb{P}$ — and $\mathbb{P}_{\mathcal{C}} = \{cc^\dagger : c \in \mathcal{C}\}$.*

Proof. For $A, B \in \mathbb{P}_{\mathcal{C}}$ the geodesic is $t \mapsto A^{1/2}(A^{-1/2}BA^{-1/2})^t A^{1/2}$ [4]. Since $\mathcal{C}$ is a closed group closed under $\dagger$, it contains, with each positive A , the one-parameter group $A^t = e^{t \log A}$ (the closure of $\{A^n\}_{n \in \mathbb{Z}}$ in $GL(n, \mathbb{C})$ contains a one-parameter subgroup through A ; equivalently, $\log A \in \text{Lie } \mathcal{C}$), in particular $A^{\pm 1/2}$, hence contains the positive matrix $M = A^{-1/2}BA^{-1/2}$ and its powers M^t , hence the whole geodesic, which consists of positive elements of $\mathcal{C}$. For the last claim: $cc^\dagger \in \mathbb{P}_{\mathcal{C}}$ for $c \in \mathcal{C}$, and conversely $A = (A^{1/2})(A^{1/2})^\dagger$ with $A^{1/2} \in \mathcal{C}$. Topological closedness of $\mathbb{P}_{\mathcal{C}} = \mathcal{C} \cap \mathbb{P}$ in $\mathbb{P}$ is immediate from closedness of $\mathcal{C}$ in $GL(n, \mathbb{C})$; unique metric projection onto closed convex subsets is standard in NPC spaces [7]. $\square$

Theorem 2.8 (Orthogonal Cartan-norm projection principle). *Let $\mathcal{C} \subset GL(n, \mathbb{C})$ be a closed subgroup, closed under conjugate transpose, with $\mathfrak{c}_{\mathcal{P}} = \{\eta = \eta^\dagger : e^{t\eta} \in \mathcal{C}\}$, and let $\xi = \xi^\dagger$. Then:*

(a) *For every $c \in \mathcal{C}$ and $\eta \in \mathfrak{c}_{\mathcal{P}}$,*

$$\left. \frac{d}{dt} \right|_{t=0} \left\| \log((e^\xi e^{t\eta})^\dagger e^\xi e^{t\eta}) \right\|_F^2 = 8 \langle \eta, \xi \rangle. \quad (12)$$

(b) *If $\xi \perp \mathfrak{c}_{\mathcal{P}}$, then for every $c \in \mathcal{C}$,*

$$\left\| \log((e^\xi c)^\dagger e^\xi c) \right\|_F \geq \|2\xi\|_F, \quad \arg \min_{c \in \mathcal{C}} \left\| \log((e^\xi c)^\dagger e^\xi c) \right\|_F = \mathcal{C} \cap U(n). \quad (13)$$

(c) *Conversely, if the tangential part $\xi_T = \text{proj}_{\mathfrak{c}_{\mathcal{P}}} \xi$ is nonzero, then $c = e$ is not a minimizer: by (12) with $\eta = -\xi_T$ the functional decreases at rate $8\|\xi_T\|^2$. Hence*

$$e \text{ minimizes } c \mapsto \left\| \log((e^\xi c)^\dagger e^\xi c) \right\|_F \text{ on } \mathcal{C} \iff \xi \perp \mathfrak{c}_{\mathcal{P}}. \quad (14)$$

Proof. By congruence invariance of the affine metric, $d(I, (e^\xi c)^\dagger e^\xi c) = d((cc^\dagger)^{-1}, e^{2\xi})$, and $(cc^\dagger)^{-1} = dd^\dagger$ with $d = c^{-\dagger} \in \mathcal{C}$, so it ranges over $\mathbb{P}_{\mathcal{C}}$ as c ranges over $\mathcal{C}$ (Lemma 2.7). For (a), take $c = e^{t\eta}$: then $(cc^\dagger)^{-1} = e^{-2t\eta}$ is a geodesic through I with velocity -2η , and the NPC first-variation formula gives $\frac{d}{dt}d(e^{-2t\eta}, e^{2\xi})^2|_0 = -2\langle -2\eta, 2\xi \rangle = 8\langle \eta, \xi \rangle$. For (b), every point of $\mathbb{P}_{\mathcal{C}}$ is joined to I by a geodesic $t \mapsto e^{2t\eta}$, $\eta \in \mathfrak{c}_{\mathfrak{p}}$, inside $\mathbb{P}_{\mathcal{C}}$; along it $h(M) = d(M, e^{2\xi})^2$ is strictly convex unless constant, and its first variation at I vanishes by orthogonality. Hence I is the unique minimizer of h on $\mathbb{P}_{\mathcal{C}}$, with value $\|2\xi\|_F^2$; the minimizing c are exactly those with $cc^\dagger = I$. Part (c) is immediate from (a). $\square$

The numerical experiments that preceded this proof (random tangential and normal directions ξ) are retained in the repository as regression checks of (12) and of both directions of (c).

Proposition 2.9 (Canonical compatible hull, stratified). *Let $\theta(g) = (g^\dagger)^{-1}$ and let $C_\theta(H)$ be the smallest closed subgroup containing $H = \text{Stab}(J, G'_0)$ and $\theta(H)$. In the canonical coordinates,*

$$C_\theta(H) = C_1 \times C_2, \quad C_j = \begin{cases} SL(2, \mathbb{C}), & \omega_j \neq 1, \\ SO(2, \mathbb{C}), & \omega_j = 1, \end{cases} \quad (15)$$

acting on the j -th mode plane. On the generic stratum the compatible hull is the full mode-preserving symplectic subgroup: not merely a convenient supergroup but the minimal θ -stable subgroup forced by the physical stabilizer. At an alignment value $\omega_j = 1$ the j -th factor of H is already θ -stable and the hull degenerates to H_j itself.

Proof. H and $\theta(H)$ are contained in the block-diagonal subgroup, so $C_\theta(H) \subseteq SL(2, \mathbb{C})^2$. For the converse it suffices to show the real Lie algebra generated by $\mathfrak{h}_j = \mathbb{C}X_j$ and $\theta(\mathfrak{h}_j) = \mathbb{C}X_j^\dagger$ is all of $\mathfrak{sl}(2, \mathbb{C})$ on each mode. On mode 1 (block coordinates), $X_1 = \begin{pmatrix} 0 & \omega_2 \\ -\omega_2^{-1} & 0 \end{pmatrix}$ and

$$[X_1, X_1^\dagger] = (\omega_2^2 - \omega_2^{-2}) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \neq 0 \quad (\omega_2 \neq 1), \quad (16)$$

and the real span of $X_1, X_1^\dagger, [X_1, X_1^\dagger]$ and their i -multiples is six-dimensional, hence equals $\mathfrak{sl}(2, \mathbb{C})$; likewise on mode 2 with ω_1 . The generated group is then the connected $SL(2, \mathbb{C})$ on each generic mode. At $\omega_j = 1$ one has $X_j^\dagger = -X_j$, so $\theta(\mathfrak{h}_j) = \mathfrak{h}_j$ and the closure is $\mathfrak{so}(2, \mathbb{C})$ itself. $\square$

(As regression evidence, not as part of the proof: the Lie-closure dimensions 6 and 2 are also confirmed computationally in the repository.)

Remark 2.10 (Alignment: hull contraction versus physical enhancement). The stratification of Proposition 2.9 is mirrored by two distinct discontinuities at $\omega_j = 1$, which must not be conflated. For the *hull*, the compact locus contracts, $C_j \cap K : SU(2) \rightarrow U(1)$, because the hull itself shrinks from $SL(2, \mathbb{C})$ to $SO(2, \mathbb{C})$; accordingly, for minimization over the hull the equality locus *changes* from $SU(2)$ -type to $U(1)$ -type. For the *physical stabilizer* the unitary locus is enhanced, $H_j \cap K : \mathbb{Z}_2 \rightarrow U(1)$, and it is only for the variational problem restricted to H that the equality locus of Theorem 2.5 *enlarges* at alignment. The enhancement is a property of the physical embedding (the twist), not of the abstract groups, and marks the parameter values at which the twisted and compatible pictures coincide.

Corollary 2.11 (Second proof of Theorem 2.5). *Apply Theorem 2.8(b) with $\mathcal{C} = C_\theta(H) = SL(2, \mathbb{C})^2$, whose positive part $\mathbb{P}_\mathcal{C} \cong \mathbb{H}^3 \times \mathbb{H}^3$ is totally geodesic in $\mathbb{P}$, the Gram data (τ_j, n_j) of Lemma 2.2 being polar coordinates on the two hyperbolic factors; and with $\xi = \frac{r}{2}B$, which is block-off-diagonal and hence orthogonal to $\mathfrak{c}_\mathfrak{p}$ (the per-mode traceless Hermitian blocks). This yields $\mathcal{F}(S_+c) \geq \mathcal{F}(S_+)$ for every c in the hull, with the two equality records kept separate:*

$$\arg \min_{c \in \mathcal{C}} \mathcal{F}(S_+c) = \mathcal{C} \cap K, \quad \arg \min_{h \in H} \mathcal{F}(S_+h) = H \cap K, \quad (17)$$

the second following from the first because $e \in H \subset \mathcal{C}$ realizes the unrestricted minimum.

Remark 2.12. (i) The closed form (8) is the Pythagorean law of the ambient symmetric space: the worst-alignment configurations sweep a totally geodesic $\mathbb{H}^2 \times \mathbb{R}$, on which $\operatorname{arccosh}(\cosh r \cosh b)$ is the hyperbolic hypotenuse law ($\cosh h = \cosh r \cosh b$, legs meeting at a right angle in $\mathbb{H}^2$) while a adds by flat Pythagoras in the $\mathbb{R}$ factor. This exact value is content beyond the projection principle, which yields only the inequality. (ii) On the compactness subtlety: $H \cap K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ refers to the intersection with the ambient maximal compact determined by the physical inner product. Abstractly $SO(2, \mathbb{C}) \cong \mathbb{C}^\times$ has maximal compact $U(1)$ after a suitable conjugation; the finiteness is a property of the physical embedding — which is precisely the point: the embedding, not the group, carries the obstruction, and Proposition 2.9 locates the minimal compatible geometry it forces.

2.2 The n -mode statement

The projection principle is dimension-independent, so the variational selection extends verbatim to any number of PT mode pairs.

Theorem 2.13 ($\operatorname{Sp}(2n, \mathbb{C})$ selection principle). *Fix a symplectic splitting of $\mathbb{C}^{2n}$ into n canonical mode planes and let $\mathcal{C}_n \cong \prod_{j=1}^n SL(2, \mathbb{C}) \subset \operatorname{Sp}(2n, \mathbb{C})$ be the mode-preserving subgroup, with positive part $\mathbb{P}_{\mathcal{C}_n} \cong (\mathbb{H}^3)^n$. The orthogonal complement of $\mathfrak{c}_{n,\mathfrak{p}}$ in the Hermitian Hamiltonian directions consists exactly of the inter-mode couplings (Hermitian Hamiltonian matrices with vanishing mode-diagonal blocks). Consequently, for every inter-mode $\xi = \xi^\dagger$,*

$$\min_{c \in \mathcal{C}_n} \left\| \log((e^\xi c)^\dagger e^\xi c) \right\|_F = \|2\xi\|_F, \quad \arg \min = \mathcal{C}_n \cap K, \quad (18)$$

and for any quadratic PT Hamiltonian whose positive normal-form diagonalizer is e^ξ with inter-mode ξ , and whose stabilizer is contained in $\mathcal{C}_n$, the positive diagonalizer is the unique minimum-distortion diagonalizer up to unitary stabilizer elements, and all minimizers define the same Hilbert-space metric.

Proof. Write $\xi \in \mathfrak{p}$ in 2×2 blocks $(\xi_{ij})_{i,j=1}^n$ relative to $V = V_1 \oplus \cdots \oplus V_n$. A one-parameter group $e^{t\eta}$ lies in $\mathcal{C}_n$ iff η is block-diagonal, so $\mathfrak{c}_{n,\mathfrak{p}} = \{\xi \in \mathfrak{p} : \xi_{ij} = 0, i \neq j\}$ (per-mode Hermitian traceless blocks: Hermiticity and the 2×2 Hamiltonian condition make each ξ_{jj} a traceless Hermitian block). Since $\operatorname{tr}(\xi\eta) = \sum_{ij} \operatorname{tr}(\xi_{ij}\eta_{ji})$ and η is block-diagonal, the pairing sees only $\sum_j \operatorname{tr}(\xi_{jj}\eta_{jj})$, so $\xi \perp \mathfrak{c}_{n,\mathfrak{p}}$ iff every diagonal block ξ_{jj} vanishes: the normal space is exactly the inter-mode couplings, whose paired off-diagonal blocks ξ_{ji} are determined from ξ_{ij} by the Hermitian and symplectic conditions. The dimension identity $n(2n+1) = 3n+2n(n-1)$ is then a check, not a step. Everything else is Theorem 2.8 applied to $\mathcal{C}_n$, plus the polar-factor argument of Theorem 2.5 for the metric statement. $\square$

For the PU oscillator, $n = 2$ and $\xi = \frac{r}{2}B$ is inter-mode; the Bender–Mannheim metric is the case in point. The model-dependent input in any application is thus reduced to a single check: that the positive diagonalizer direction is inter-mode for the chosen splitting, i.e., trace-orthogonal to $\mathfrak{c}_{n,p}$ — and by Theorem 2.8(c) this check is not only sufficient but necessary. We emphasize that the canonical-hull computation of Proposition 2.9 is proved here for the two-mode PU stabilizer; for a general n -mode system the containment of the physical stabilizer in $\mathcal{C}_n$, and the structure of its compatible hull across mixed exceptional strata, are hypotheses to be verified per system.

3 The free fourth-order field

3.1 Modewise data and the phase-space Hilbert scale

For $(\square + m_1^2)(\square + m_2^2)\phi = 0$, each spatial momentum k carries a PU pair with $\omega_i(k) = \sqrt{k^2 + m_i^2}$ and $m_1 > m_2 > 0$. Since $(\omega_1 - \omega_2)(\omega_1 + \omega_2) = m_1^2 - m_2^2$ identically, the rapidity obeys (3) *exactly*, interpolating between $r(0) = \log \frac{m_1 + m_2}{m_1 - m_2}$ and $r(k) = 2 \log |k| + O(1)$.

Corollary 3.1 (Phase-space Hilbert scale). *In the canonically normalized modewise coordinates, the observable-space metric $M_{\text{obs}}(k) = S_+(k)^\dagger S_+(k)$ has eigenvalues $e^{\pm r(k)}$ with*

$$e^{r(k)} = \frac{(\omega_1(k) + \omega_2(k))^2}{m_1^2 - m_2^2} \asymp 1 + k^2, \quad e^{-r(k)} \asymp (1 + k^2)^{-1}. \quad (19)$$

On the spectral subspaces $\Pi_\pm(k)$ of B the induced one-particle phase-space norm is therefore equivalent to an $H^1 \oplus H^{-1}$ Sobolev pair (up to the constant $m_1^2 - m_2^2$ and bounded factors): it is not uniformly equivalent to the standard norm, and no scalar renormalization makes it so.

This is the precise content of the observation that the modewise metric “blows up in the UV”. It concerns the classical (phase-space) metric. The quantum state-level comparison is finer, and is computed next.

3.2 The PT ground state, exactly

Fix a mode and work in the normalized coordinates (x, y) with $\omega_1 > \omega_2 > 0$. The positive Hamiltonian $\tilde{H}'$ has ground state $\phi_0 \propto e^{-\frac{1}{2}(x^2/\omega_2 + \omega_1 y^2)}$. The PT Hamiltonian in these coordinates is the differential operator

$$H'_{\text{PT}} = \frac{1}{2} \left[\frac{\omega_1^2 + \omega_2^2}{\omega_1 \omega_2} x^2 + \omega_1 \omega_2 y^2 - \omega_1 \omega_2 \partial_x^2 \right] - x \partial_y. \quad (20)$$

Theorem 3.2 (PT ground state). *The unique Gaussian solution of the modewise annihilation equations $(\ell_j^\top S_+^{-1} V) \psi_0 = 0$ is*

$$\psi_0(x, y) \propto \exp \left[-\frac{1}{2} \left(\frac{\omega_1 + \omega_2}{\omega_1 \omega_2} x^2 + 2xy + (\omega_1 + \omega_2) y^2 \right) \right], \quad (21)$$

a real Gaussian with $\det(\text{quadratic form}) = \frac{\omega_1^2 + \omega_1 \omega_2 + \omega_2^2}{\omega_1 \omega_2} > 0$: normalizable for all $\omega_1 > \omega_2 > 0$, and $H'_{\text{PT}} \psi_0 = \frac{\omega_1 + \omega_2}{2} \psi_0$. Its fidelity with the two-oscillator vacuum and its occupation relative

to it are exactly

$$f(\omega_1, \omega_2) = |\langle \hat{\phi}_0 | \hat{\psi}_0 \rangle|^2 = \frac{4\omega_1 \sqrt{\omega_1^2 + \omega_1\omega_2 + \omega_2^2}}{4\omega_1^2 + 3\omega_1\omega_2 + \omega_2^2}, \quad (22)$$

$$\langle N \rangle(\omega_1, \omega_2) = \langle \hat{\psi}_0 | (N_1 + N_2) | \hat{\psi}_0 \rangle = \frac{\omega_2(\omega_1^2 + \omega_2^2)}{2\omega_1(\omega_1^2 + \omega_1\omega_2 + \omega_2^2)}, \quad \langle N_1 \rangle = \langle N_2 \rangle. \quad (23)$$

In particular, along the field dispersion $\omega_i = \omega_i(k)$,

$$\lim_{|k| \rightarrow \infty} f = \frac{\sqrt{3}}{2} \approx 0.866, \quad \lim_{|k| \rightarrow \infty} \langle N \rangle = \frac{1}{3}, \quad (24)$$

universal constants independent of m_1, m_2, γ .

Proof. Solving the two first-order equations for a Gaussian ansatz gives the stated quadratic form (one solution); the eigenvalue equation is a direct substitution. The fidelity is the standard two-Gaussian determinant formula; the occupations follow from covariance algebra, $\langle N_j \rangle = \frac{1}{2} c_j^\top \Sigma c_j$ with Σ the covariance of $|\psi_0|^2$ and c_j the coefficient vectors of $a_j \psi_0$. The limits follow with $\omega_1/\omega_2 \rightarrow 1$. (Independent symbolic verification of each step, and a 25-digit quadrature check of the fidelity, are recorded in the repository as verification evidence.) $\square$

Remark 3.3. Two features deserve emphasis. (i) ψ_0 is a smooth, normalizable Gaussian even as $\omega_1 \rightarrow \omega_2$: the exceptional-point pathology (divergence of S_+ , Jordan structure) lives in the similarity transformation and the excited spectrum, not in the ground state itself. (ii) A naive Bogoliubov estimate would assign the modewise map a squeezing $\sim \sinh^2(r(k)/2) \sim k^2$ and hence a violently divergent particle content per mode. The exact result (24) is $O(1)$: the diverging rapidity $r(k)$ is carried almost entirely by directions that do not create quanta. The naive estimate fails because $\rho = e^{-Q/2}$ is not $*$ -preserving on the original Fock structure — the transformed “annihilators” $\rho a_j \rho^{-1}$ do not satisfy the unitary Bogoliubov normalization ($\alpha \alpha^\dagger - \beta \beta^\dagger = \cosh r \neq 1$) — so Shale–Stinespring-type criteria cannot be applied to ρ directly. Comparing the two *states*, both of which live in the same L^2 , avoids this trap entirely.

3.3 Disjointness of the representations

Theorem 3.4 (Representation change). *Let $m_1 > m_2 > 0$, $d \geq 1$, and consider the free fourth-order field in a box of volume V with momentum cutoff Λ , with $\Psi_{\text{PT}} = \bigotimes_k \hat{\psi}_0(k)$ the modewise PT ground state and $\Omega = \bigotimes_k \hat{\phi}_0(k)$ the free two-field Fock vacuum. Then:*

1. *the vacuum overlap satisfies $|\langle \Omega | \Psi_{\text{PT}} \rangle|^2 = \prod_k f(k) \leq e^{-c V \Lambda^d}$ for some $c > 0$ depending only on m_1/m_2 and d (since $f(k) \rightarrow \sqrt{3}/2 < 1$);*
2. *the relative particle density diverges in the UV: $\frac{1}{V} \langle N_{\text{tot}} \rangle = \int^\Lambda \frac{d^d k}{(2\pi)^d} \langle N \rangle(k) \sim \frac{1}{3} \int^\Lambda \frac{d^d k}{(2\pi)^d} = \infty$ as $\Lambda \rightarrow \infty$, in every dimension $d \geq 1$;*
3. *in the thermodynamic/continuum limit the incomplete-tensor-product classes of Ψ_{PT} and Ω differ ($\sum_k (1 - |\langle \hat{\psi}_0 | \hat{\phi}_0 \rangle|) = \infty$), so the GNS representations of the CCR algebra built on the two states are disjoint: no density matrix in the Fock representation of the free pair (m_1, m_2) reproduces the PT vacuum, and vice versa.*

Each mode, by contrast, is unitarily equivalent: the modewise Dyson maps $\rho_k = e^{-Q_k/2}$ define unitaries between the mode Hilbert spaces equipped with the physical (η_k) and standard inner products.

Proof. (1) and (2) are immediate from Theorem 3.2: $f(k)$ and $\langle N \rangle(k)$ are continuous in k with the universal limits (24), so beyond some $k_*(m_1/m_2)$ one has, e.g., $f(k) \leq 0.9$ and $\langle N \rangle(k) \geq 0.3$. (3) is von Neumann’s equivalence criterion for incomplete tensor products [8] applied to the two product vectors, combined with purity: inequivalent pure product states generate disjoint representations. The modewise statement is the finite-dimensional similarity of [3, 1]. $\square$

Remark 3.5 (What this does and does not say). The theorem does not say the PT quantization of the fourth-order field is inconsistent. The physical Hilbert space, defined as the completion of the modewise-unitarized theory, is naturally identified with the Fock space of the free pair (m_1, m_2) ; on it the dynamics is unitary, exactly as argued by Bender and Mannheim modewise. What fails is the *global* equivalence between this physical representation and the naive covariant Fock representation in which the fourth-order propagator and perturbation theory are usually written: the two are different, mutually singular representations of the same field algebra, with an $O(1)$ discrepancy per UV mode — not a discrepancy that decays with k , and not one confined to the exceptional locus $m_1 = m_2$. Any interacting construction therefore has to choose a representation first; formal manipulations that mix them (e.g., inserting $\eta = e^{-Q}$ as if it were a bounded operator on the covariant Fock space) are not supported: by Corollary 3.1 the metric is unbounded with unbounded inverse, defining a Hilbert scale rather than an equivalent norm, and by Theorem 3.4 its formal infinite-mode product annihilates the covariant vacuum’s representation class entirely. This sharpens, for the free theory, the sense in which PT-based unitarity claims for quadratic gravity [9] require an explicit domain-and-completion construction.

4 Discussion

The variational principle. Theorem 2.5 upgrades the uniqueness theorem of [1]: the Bender–Mannheim metric is not merely the unique positive point of the diagonalizer coset but its unique distortion minimizer, with the exact excess $4[a^2 + \operatorname{arccosh}^2(\cosh r \cosh b)] - 4r^2$ in the Gram rapidities of the stabilizer. Section 2.1 shows this is an instance of an orthogonal Cartan-norm projection principle along $\dagger$ -closed subgroups — classical in spirit [5, 6], here proved in if-and-only-if form with the minimal compatible hull identified canonically (Proposition 2.9) — while the elementary proof supplies the minimizing value in closed form, which the abstract theory does not. The n -mode version (Theorem 2.13) makes the portability to general quadratic PT systems a theorem rather than a claim.

The two geometries, again. The field-theoretic analysis separates two statements that are easily conflated. The *phase-space* metric degenerates in the UV exactly as the rapidity formula (3) dictates ($e^{\pm r(k)} \asymp k^{\pm 2}$, a Sobolev pair $H^1 \oplus H^{-1}$); this is a statement about observables. The *state-space* mismatch saturates at universal constants $(\sqrt{3}/2, 1/3)$; this is a statement about representations, and it is what controls implementability. Both are exact consequences of the same modewise data. The constants are universal because the UV limit $\omega_1/\omega_2 \rightarrow 1$

erases all mass and coupling dependence — remarkably, the same limit in which the finite-dimensional theory degenerates (the exceptional point) is the limit in which the ground-state data become featureless.

Outlook. Three natural continuations: (i) sharpen Theorem 2.13 into a classification: for which quadratic PT Hamiltonians is the positive diagonalizer direction inter-mode for some canonical splitting (by Theorem 2.8(c) this is necessary and sufficient), and what replaces the selection principle when it fails; (ii) determine whether the disjointness survives interactions in low dimensions, where the free-theory obstruction is only the universal 1/3 per mode (a wave-function renormalization of the modewise map, $\rho_k \rightarrow \rho_k Z_k$ with Z_k unitary, cannot change class (3), but a k -dependent redefinition of the mode split might); and (iii) make contact with the Krein-space formulation of indefinite-metric QFT, where the covariant representation lives, to phrase the representation choice intrinsically. We leave these to future work.

References

- [1] *Canonical positive symplectic diagonalization of the Pais–Uhlenbeck oscillator*, companion paper and verification repository, 2026.
- [2] C. M. Bender and P. D. Mannheim, Phys. Rev. Lett. **100** (2008) 110402.
- [3] C. M. Bender and P. D. Mannheim, Phys. Rev. D **78** (2008) 025022.
- [4] R. Bhatia, *Positive Definite Matrices*, Princeton University Press, 2007.
- [5] G. D. Mostow, *Self-adjoint groups*, Ann. of Math. **62** (1955) 44.
- [6] G. Kempf and L. Ness, *The length of vectors in representation spaces*, in: Algebraic Geometry, Lecture Notes in Math. **732**, Springer, 1979, 233.
- [7] M. R. Bridson and A. Haefliger, *Metric Spaces of Non-Positive Curvature*, Springer, 1999.
- [8] J. von Neumann, *On infinite direct products*, Compos. Math. **6** (1939) 1.
- [9] J. Kubo and J. Kuntz, *Unitarity through PT symmetry in quantum quadratic gravity*, arXiv:2410.08278.